

Your grade: 100%

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Next item →

1. A discrete-time system is described by the (discrete-time) matrices:

1 / 1 point

$A = \begin{bmatrix} 0.1 & 0.2 \\ 0.3 & 0.4 \end{bmatrix}, B = \begin{bmatrix} 0.5 \\ 0.6 \end{bmatrix}, C = \begin{bmatrix} 0.7 & 0.8 \end{bmatrix},$ and $D = 0.9$.

What is its observability matrix?

- ☒ $\mathcal{O} = \begin{bmatrix} 0.7 & 0.8 \\ 0.31 & 0.46 \end{bmatrix}.$
- ☐ $\mathcal{O} = \begin{bmatrix} 0.7 & 0.31 \\ 0.8 & 0.46 \end{bmatrix}.$
- ☐ $\mathcal{O} = \begin{bmatrix} 0.5 & 0.17 \\ 0.6 & 0.39 \end{bmatrix}.$
- ☐ $\mathcal{O} = \begin{bmatrix} 0.5 & 0.6 \\ 0.17 & 0.39 \end{bmatrix}.$

✔ Correct
Yes. This is correct.

2. A discrete-time system is described by the (discrete-time) matrices:

1 / 1 point

$A = \begin{bmatrix} 0.1 & 0.2 \\ 0.3 & 0.4 \end{bmatrix}, B = \begin{bmatrix} 0.5 \\ 0.6 \end{bmatrix}, C = \begin{bmatrix} 0.7 & 0.8 \end{bmatrix},$ and $D = 0.9$.

What is its controllability matrix?

- ☐ $\mathcal{C} = \begin{bmatrix} 0.7 & 0.8 \\ 0.31 & 0.46 \end{bmatrix}.$
- ☐ $\mathcal{C} = \begin{bmatrix} 0.7 & 0.31 \\ 0.8 & 0.46 \end{bmatrix}.$
- ☒ $\mathcal{C} = \begin{bmatrix} 0.5 & 0.17 \\ 0.6 & 0.39 \end{bmatrix}.$
- ☐ $\mathcal{C} = \begin{bmatrix} 0.5 & 0.6 \\ 0.17 & 0.39 \end{bmatrix}.$

✔ Correct
Yes. This is correct.

3. A discrete-time system is described by the (discrete-time) matrices:

1 / 1 point

$A = \begin{bmatrix} 0.1 & 0 \\ 0 & 0.2 \end{bmatrix}, B = \begin{bmatrix} 0.3 \\ 0.4 \end{bmatrix}, C = \begin{bmatrix} 0.5 & 0.0 \end{bmatrix},$ and $D = 0.6$.

Which of the following is true?

- ☐ The system is both controllable and observable.
- ☐ The system is neither controllable nor observable.
- ☒ The system is controllable but not observable.
- ☐ The system is observable but not controllable.

✔ Correct
Yes. This is correct.